4. ANALYZE THE MALICIOUS NETWORK TRAFFIC USING WIRESHARK

DATE: 18/02/2026

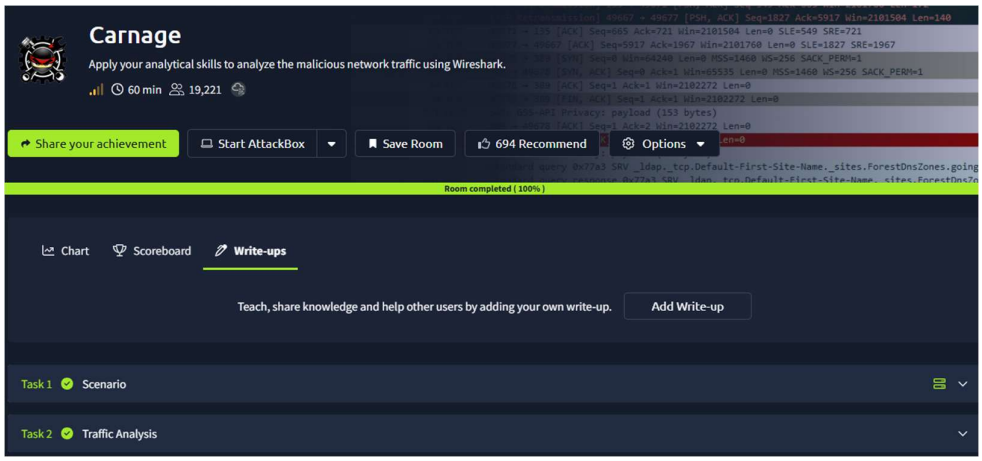
Aim:

To apply your analytical skills to analyze the malicious network traffic using Wireshark.

Procedure:

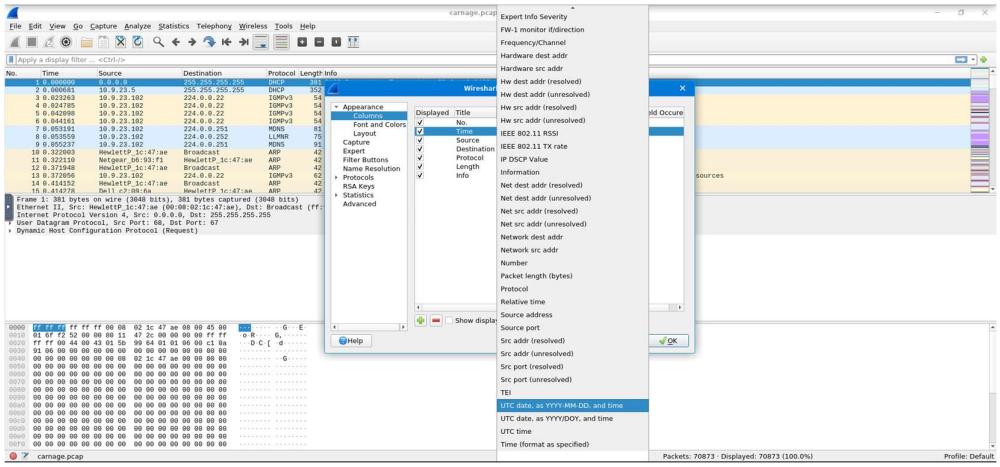
1. Log into TryHackMe and open the Network Security Essentials room.
2. Study basic network security concepts including firewalls, IDS, and IPS.
3. Understand common network threats such as malware, phishing, and DDoS attacks.
4. Learn monitoring techniques using logs and traffic analysis.
5. Explore methods to protect networks from adversaries.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

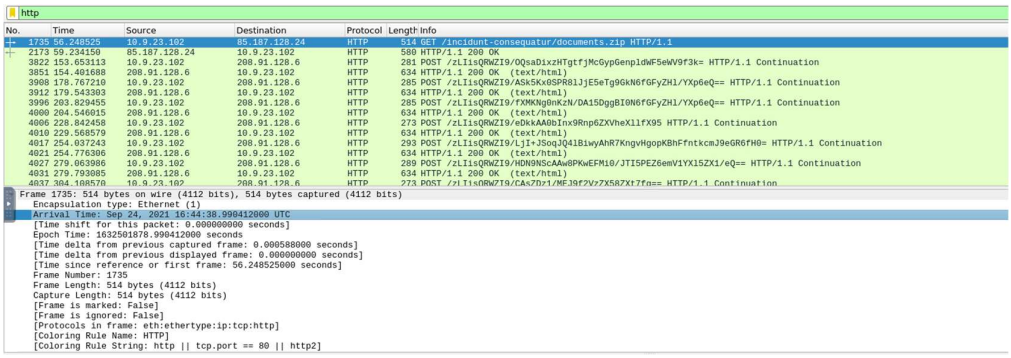
Tasks:

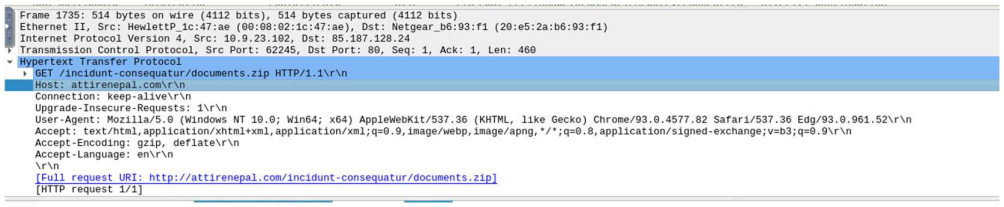


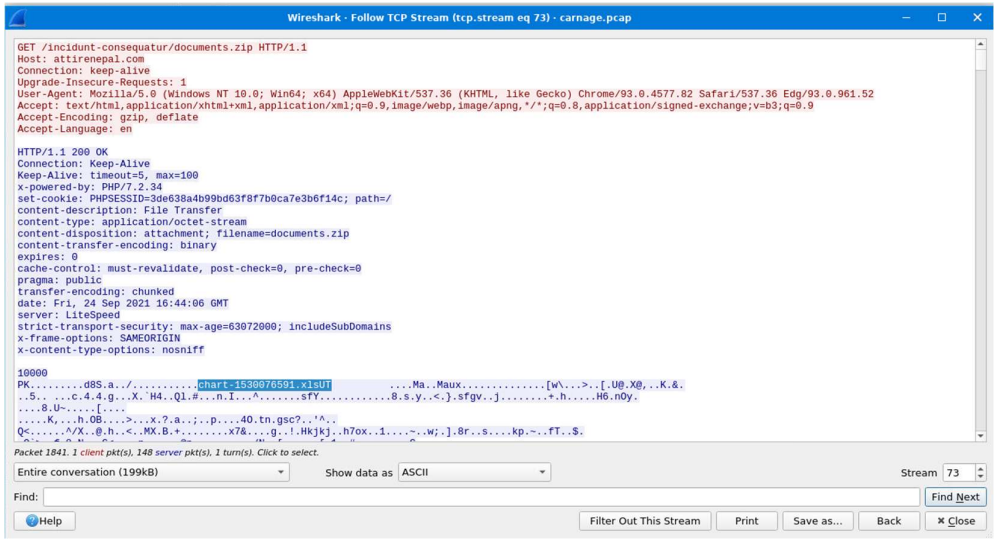
TASK 1 – Scenario

TASK 2 – Traffic Analysis









Wireshark · Follow TCP Stream (tcp.stream eq 73) · carnage.pcap

```
GET /incidunt-consequatur/documents.zip HTTP/1.1
Host: attirenepal.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/93.0.4577.82 Safari/537.36 Edg/93.0.961.52
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en

HTTP/1.1 200 OK
Connection: Keep-Alive
Keep-Alive: timeout=5, max=100
x-powered-by: PHP/7.2.34
set-cookie: PHPSESSID=3de638a4b99bd63f8f7b0ca7e3b6f14c; path=/
content-description: File Transfer
content-type: application/octet-stream
content-disposition: attachment; filename=documents.zip
content-transfer-encoding: binary
expires: 0
cache-control: must-revalidate, post-check=0, pre-check=0
pragma: public
transfer-encoding: chunked
date: Fri, 24 Sep 2021 16:44:06 GMT
server: LiteSpeed
strict-transport-security: max-age=63072000; includeSubDomains
x-frame-options: SAMEORIGIN
x-content-type-options: nosniff
```

Wireshark · Follow TCP Stream (tcp.stream eq 73) · carnage.pcap

```
GET /incidunt-consequatur/documents.zip HTTP/1.1
Host: attirenepal.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/93.0.4577.82 Safari/537.36 Edg/93.0.961.52
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en

HTTP/1.1 200 OK
Connection: Keep-Alive
Keep-Alive: timeout=5, max=100
x-powered-by: PHP/7.2.34
```

dns&&(frame.time >= "2021-09-24 16:45:11" && frame.time <= "2021-09-24 16:45:30")

No.	Time	Source	Destination	Protoc	Length	Info
2422	2021-09-24 16:45:11.440580	10.9.23.102	10.9.23.5	DNS	77	Standard query 0x04b5 A finejewels.com.au
2423	2021-09-24 16:45:11.440883	10.9.23.5	10.9.23.102	DNS	93	Standard query response 0x04b5 A finejewels.com.au A 148.72.192.206
2612	2021-09-24 16:45:16.913610	10.9.23.102	10.9.23.5	DNS	90	Standard query 0xf3e8 A self.events.data.microsoft.com
2613	2021-09-24 16:45:17.059814	10.9.23.5	10.9.23.102	DNS	212	Standard query response 0xf3e8 A self.events.data.microsoft.com CNAME se
2877	2021-09-24 16:45:20.262640	10.9.23.102	10.9.23.5	DNS	128	Standard query 0x68ce SRV _ldap._tcp.Default-First-Site-Name._sites.dc._
2878	2021-09-24 16:45:20.262991	10.9.23.5	10.9.23.102	DNS	196	Standard query response 0x68ce SRV _ldap._tcp.Default-First-Site-Name._s
2983	2021-09-24 16:45:20.487123	10.9.23.102	10.9.23.5	DNS	74	Standard query 0xa024 A thietbiagt.com
2998	2021-09-24 16:45:20.955042	10.9.23.5	10.9.23.102	DNS	90	Standard query response 0xa024 A thietbiagt.com A 210.245.90.247
3224	2021-09-24 16:45:25.457503	10.9.23.102	10.9.23.5	DNS	77	Standard query 0x65b4 A new.americold.com

Answer the questions below

What was the date and time for the first HTTP connection to the malicious IP?

(**answer format**: yyyy-mm-dd hh:mm:ss)

2021-09-24 16:44:38 — ✓ Correct Answer

What is the name of the zip file that was downloaded?

documents.zip — ✓ Correct Answer

What was the domain hosting the malicious zip file?

attirenepal.com — ✓ Correct Answer

Without downloading the file, what is the name of the file in the zip file?

chart-1530076591.xls — ✓ Correct Answer

What is the name of the webserver of the malicious IP from which the zip file was downloaded?

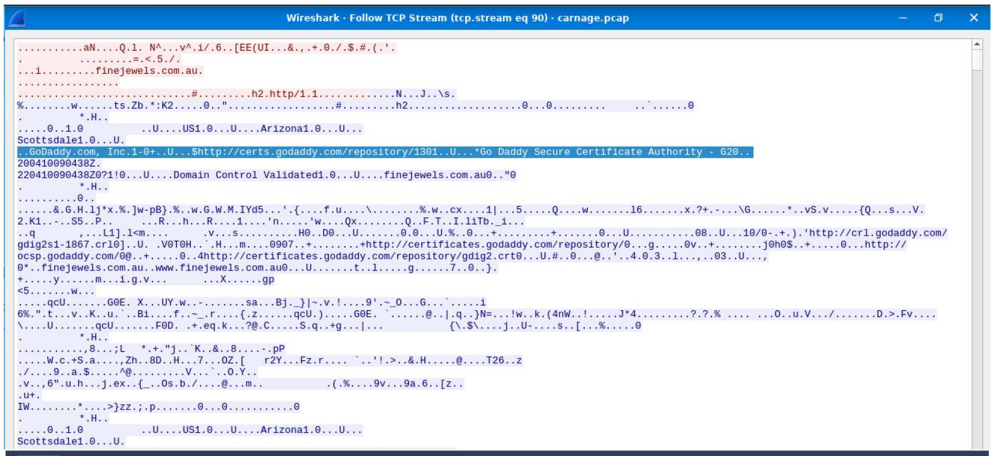
LiteSpeed — ✓ Correct Answer

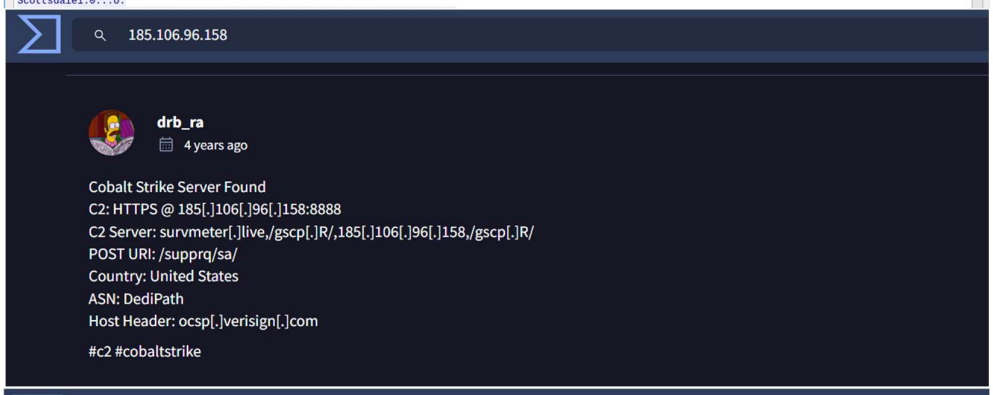
What is the version of the webserver from the previous question?

PHP/7.2.34 — ✓ Correct Answer

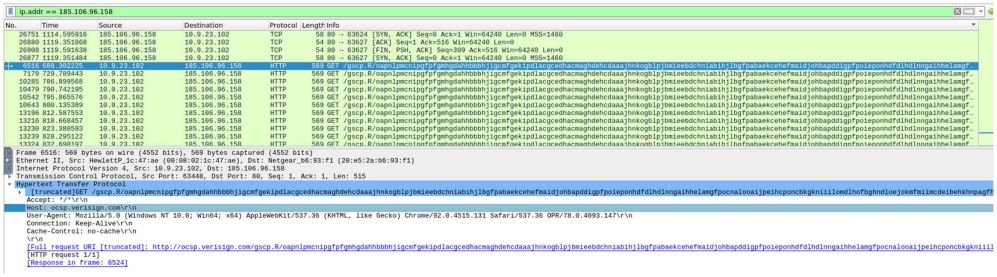
Malicious files were downloaded to the victim host from multiple domains. What were the three domains involved with this activity?

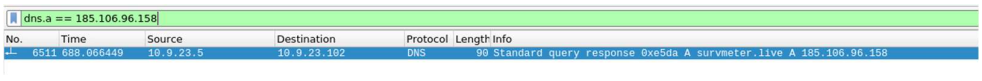
finejewels.com.au, thietbiagt.com, new.americold.com — ✓ Correct Answer

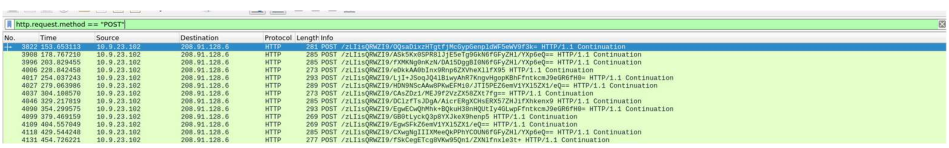

Wireshark · Follow TCP Stream (tcp.stream eq 90) · carnage.pcap

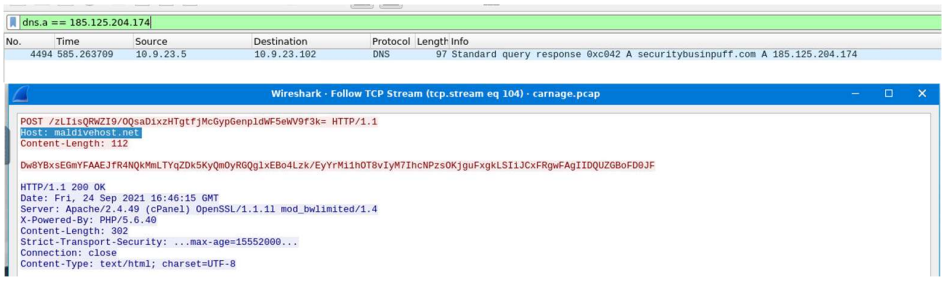

185.106.96.158
drb_ra
4 years ago
Cobalt Strike Server Found
C2: HTTPS @ 185[.]106[.]96[.]158:8888
C2 Server: survmeter[.]live,/gscp[.]R/,185[.]106[.]96[.]158,/gscp[.]R/
POST URI: /supprq/sa/
Country: United States
ASN: DediPath
Host Header: ocsp[.]verisign[.]com
#c2 #cobaltstrike

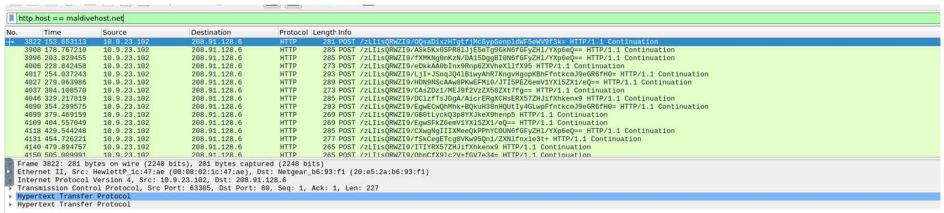
185.125.204.174
drb_ra
4 years ago
Cobalt Strike Server Found
C2: HTTPS @ 185[.]125[.]204[.]174:4444
C2 Server: securitybusinpuff[.]com,/jquery-3[.]3[.]1[.]min[.]js,185[.]125[.]204[.]174,/jquery-3[.]3[.]1[.]min[.]js
POST URI: /jquery-3[.]3[.]2[.]min[.]js
Country: N/A
ASN: Hydra Communications Ltd
#c2 #cobaltstrike

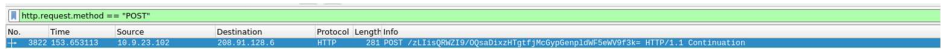

ip.addr == 185.106.96.158

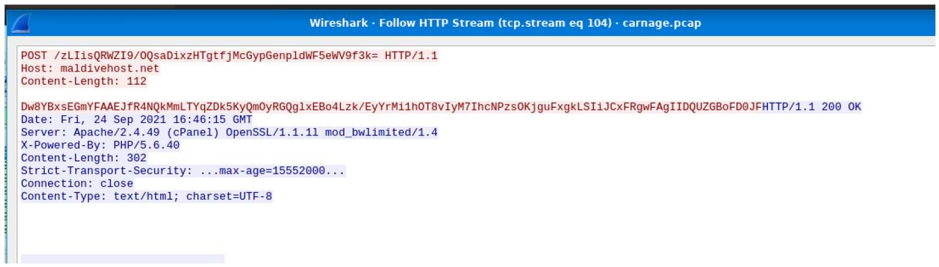

dns.a == 185.106.96.158
No. Time Source Destination Protocol Length Info
6511 688.066449 10.9.23.5 10.9.23.102 DNS 90 Standard query response 0xe5da A survmeter.live A 185.106.96.158


http.request.method == "POST"









Which certificate authority issued the SSL certificate to the first domain from the previous question?

GoDaddy — ✓ Correct Answer

What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike C2 servers. (answer format: enter the IP addresses in sequential order)

185.106.96.158, 185.125.204.174 — ✓ Correct Answer

What is the Host header for the first Cobalt Strike IP address from the previous question?

ocsp.verisign.com — ✓ Correct Answer

What is the domain name for the first IP address of the Cobalt Strike server? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

survmeter.live — ✓ Correct Answer

What is the domain name of the second Cobalt Strike server IP? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

securitybusinpuff.com — ✓ Correct Answer

What is the domain name of the post-infection traffic?

maldivehost.net — ✓ Correct Answer

What are the first eleven characters that the victim host sends out to the malicious domain involved in the post-infection traffic?

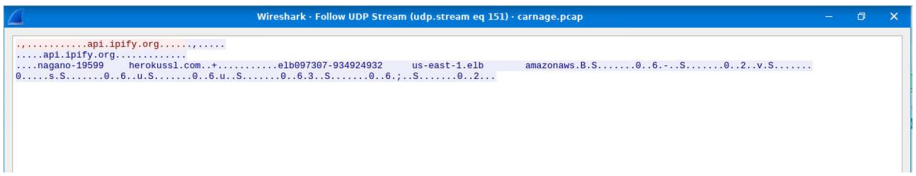
zLIisQRWZI9 — ✓ Correct Answer

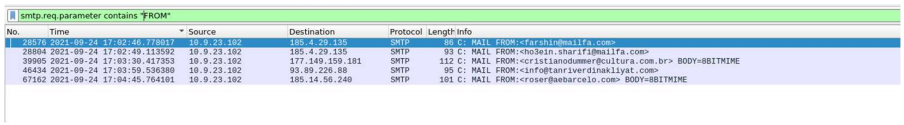
What was the length for the first packet sent out to the C2 server?

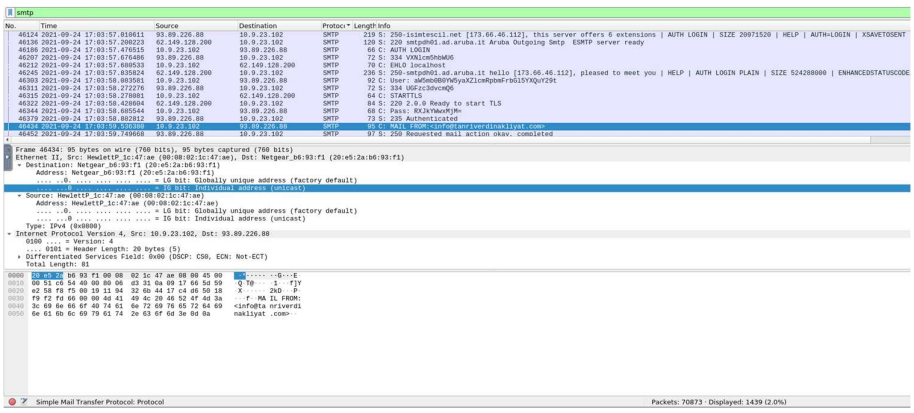
281 — ✓ Correct Answer

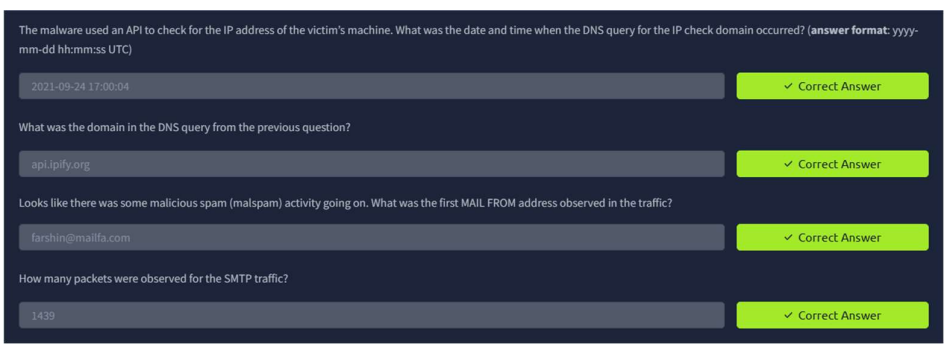
What was the Server header for the malicious domain from the previous question?

Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4 — ✓ Correct Answer









Result:

This the TryHackMe room “Carnage” was completed successfully.